Machine Learning Fundamentals

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This exercise sheet will not be graded, but it must be completed in order to pass the course.

Name: ______________________________

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Institution: ______________________________

Question 1. 3 P.

What are the two AI approaches we discussed? Describe them briefly.

Question 2. 1 P.

What is the relation between the field of *artificial intelligence*, *machine learning* and *deep learning*.

Question 3. 3 P.

What is an artificial neuron? How does it work?

Question 4. 2 P.

What is the purpose of the *loss function* in the case of *supervised learning*?

Question 5. 4 P.

Warren S. McCulloch and Walter H. Pitts Jr formalized the behavior of a neuron for the first time in 1943 using the following equation:

$$f(x_1, \ldots, x_n) = \begin{cases} 1 & \text{if } \sum_{k=1}^{n} x_k > 1 \\ 0 & \text{otherwise.} \end{cases}$$

Describe the logic of this formula in your own words. Essentially, explain how the first model of an artificial neuron works.

Question 6. 2 P.

Describe how weights in a neural network are updated during training. You can use mathematical or common language.

Question 7. 2 P.

What is the difference between a *regression* and a *classification task*? Name one example for each type of task.

Question 8. 3 P.

Define the **Mean Squared Error (MSE)** and explain how it is used in regression tasks. You can use mathematical or common language.

Question 9. 1 P.

Draw a schema that implement the logical function AND with artificial neurons. **Hint:** The neuron needs two inputs and one output which is 1 if and only if both inputs are 1.

Question 10. 3 P.

Given the rule of Modus Ponens, that is, if $R \to W$ and R is true it follows that W is true or more formally:

$$R \to W, R \vdash W.$$

Explain how a computer can calculate something we call conclusion and why this is realized by symbolic manipulation. How do these symbols get their meaning?

Question 11. 5 P.

Compare **discriminative** and **generative** models. Provide one example of each.

Question 12. 2 P.

Argue why **generative models** are usually more computationally expensive to generate a result than **discriminative models**.

Question 13. 3 P.

Why should we use a validation set in machine learning?

Question 14. 3 P.

Why should we use a test set in machine learning? **Hint**: We use the validation set for tuning our model.

Question 15. 3 P.

Explain the terms *batch*, *epoch*, and *learning rate*?

Question 16. 3 P.

Transformers-based architectures power currently the most powerful generative models. Search for a simple task they struggle with and try to understand why they struggle.

Question	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	Total
Points	3	1	3	2	4	2	2	3	1	3	5	2	3	3	3	3	43
Reached																	